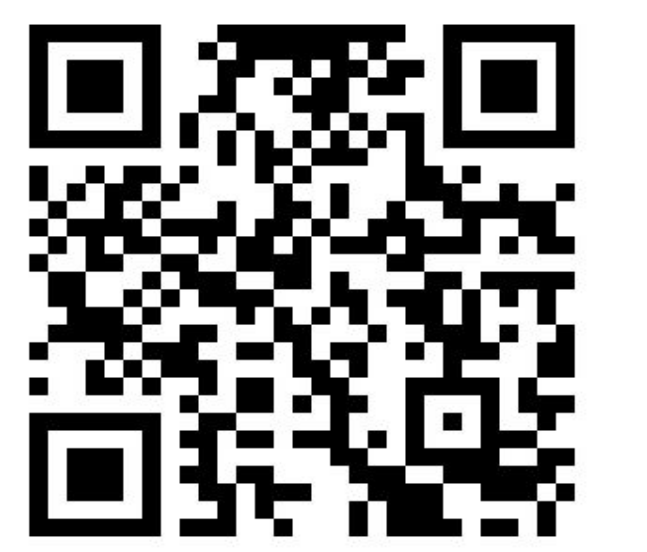


Accelerating Quantitative Feature Engineering - A C++20/pybind11 Extension for a Python-Based Trading Research Platform

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2026
September 12-18
Aurora, Colorado, USA



AEQUITAS Website

NC STATE UNIVERSITY

ABSTRACT. AEQUITAS is a **Python**-based quantitative research platform whose primary computational bottleneck consists of **19 rolling-window features** derived from OHLCV market data. This work presents a **C++20** reimplementaion of these kernels, integrated with **Python** through **pybind11** using **zero-copy NumPy buffers** and explicit **Global Interpreter Lock (GIL)** release around compute-intensive loops. All kernels reproduce the pandas reference implementation within a maximum numerical deviation of 1.1×10^{-8} . At one million rows, per-kernel speedups range from **1.3x** for exponentially weighted moving calculations, where pandas already relies on optimized compiled routines, to **27.4x** for **Average True Range (ATR)**. For a parallel **RSI-14** workload comprising eight symbols with one million rows each, execution using eight **Python threads** completes in **5.2 ms**, compared with **197.5 ms** for sequential pandas, yielding a **37.7x** speedup. The complete **19-feature** pipeline achieves a **2.1x** overall speedup, with the remaining runtime dominated by DataFrame construction and assembly rather than numerical computation.

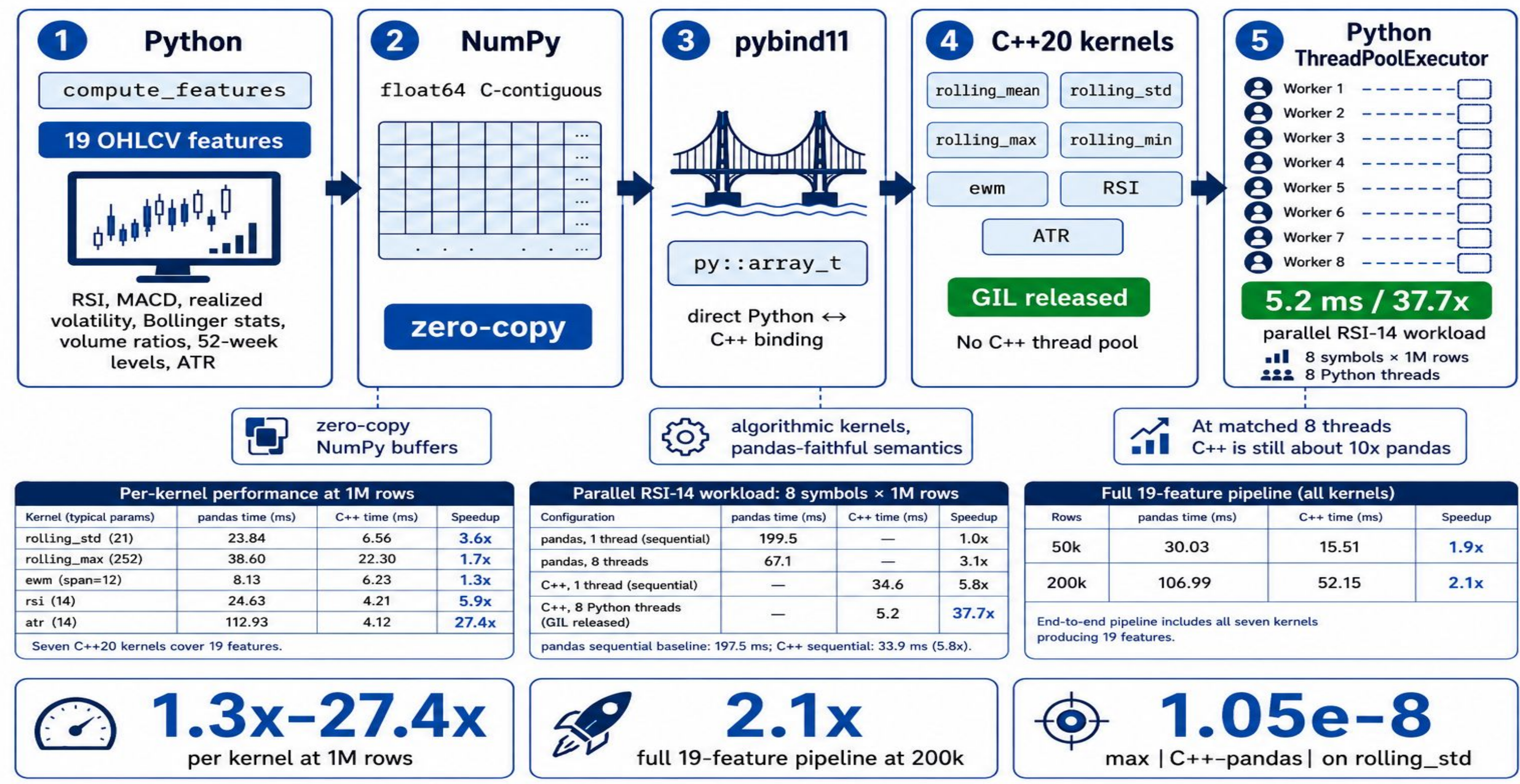
The Bottleneck

Python is a widely used language in quantitative research, while performance-critical workloads are often implemented in **C++**, as seen in libraries such as **NumPy**, **pandas**, and **PyTorch**. **AEQUITAS** is an open trading-research platform whose feature-engineering pipeline computes **19** technical and statistical indicators from **OHLCV** data, including **RSI**, **MACD**, **realized volatility**, **Bollinger statistics**, **volume ratios**, **52-week levels**, and **ATR**. This work presents a **C++20** retrofit of these kernels, exposed to **Python** through **pybind11** with zero-copy data access. The implementation preserves numerical equivalence with the original pandas pipeline, including cases requiring careful handling of rolling-window and boundary semantics.

Seven Kernels

Kernel	Algorithm	Pandas Equivalent
rolling_mean/std	Sliding sum (+ sum of squares), O(n)	rolling().mean / .std
rolling_max / min	Monotonic deque, O(n) amortized	rolling().max / .min
ewm_mean	Recurrence, adjust=False	ewm(adjust=False).mean
rsi	Wilder smoothing, $\alpha = 1/n$	_rsi() in features.py
atr	True range + EMA(span=n)	_atr() in features.py

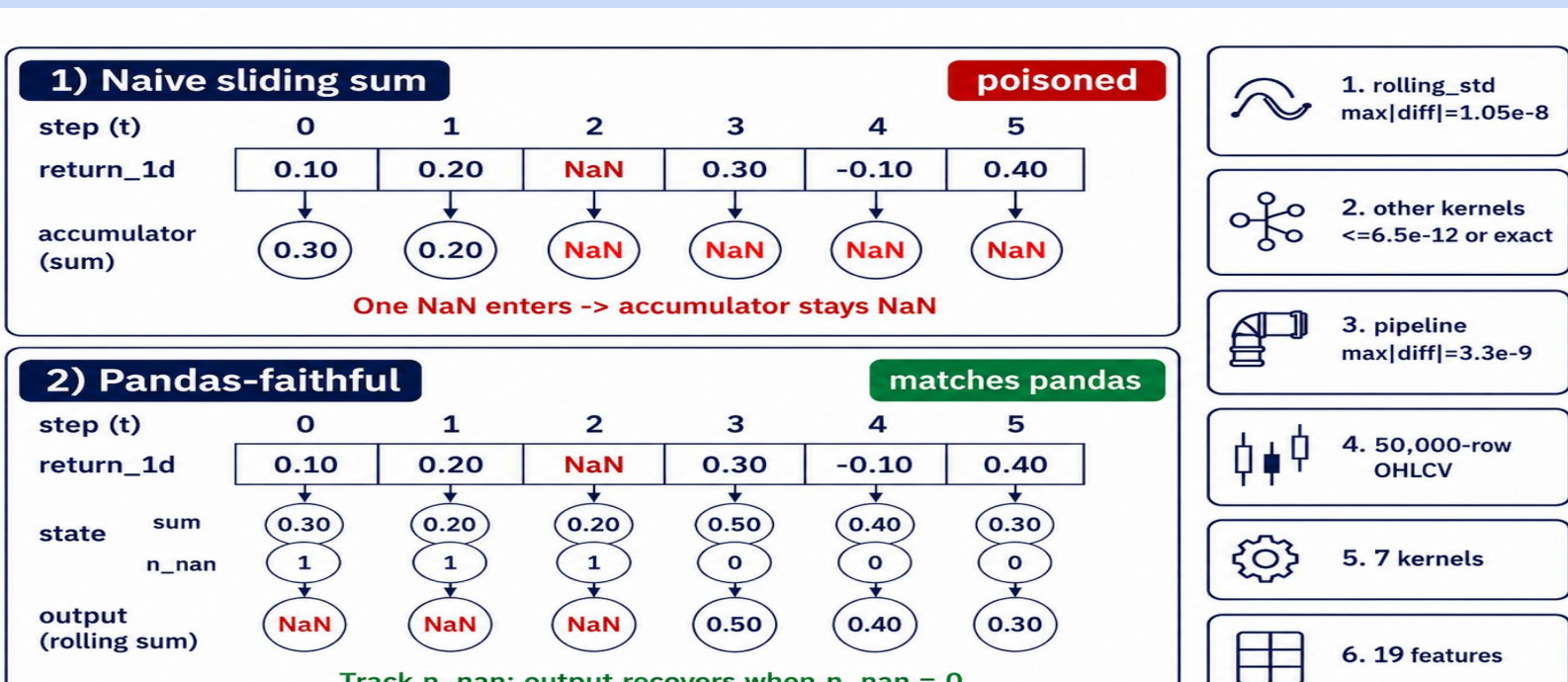
Pipeline Architecture



GIL snippet

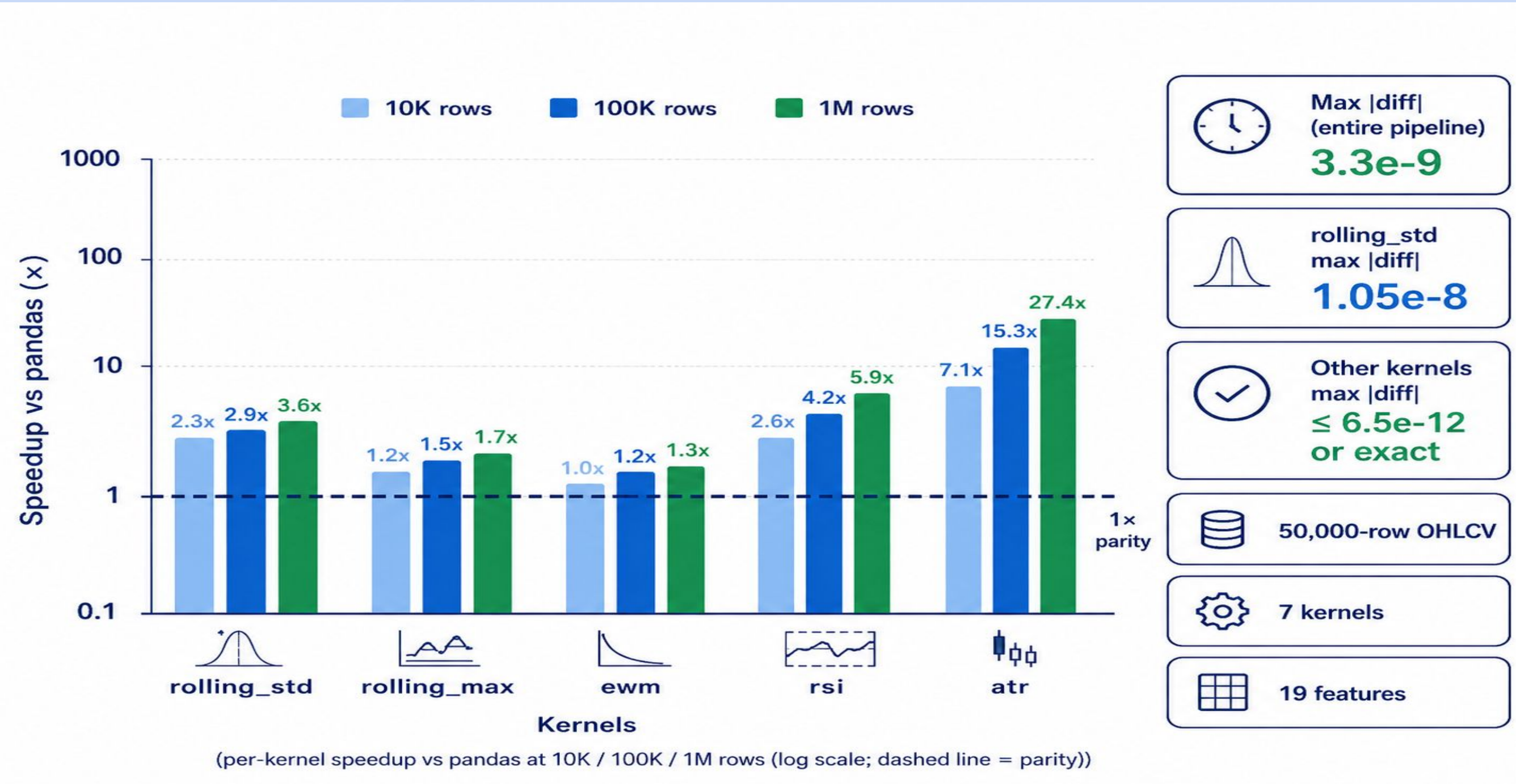
```
// GIL Release and Parallel Compute Snippet
Arr rsi(Arr close, int period) {
    auto s = in(close); double* optr;
    Arr out = make_out(s.n, &optr);
    {
        py::gil_scoped_release rel;
        rsi_impl(s.p, optr, s.n, period);
    }
    return out;
}
```

NaN Edge Case

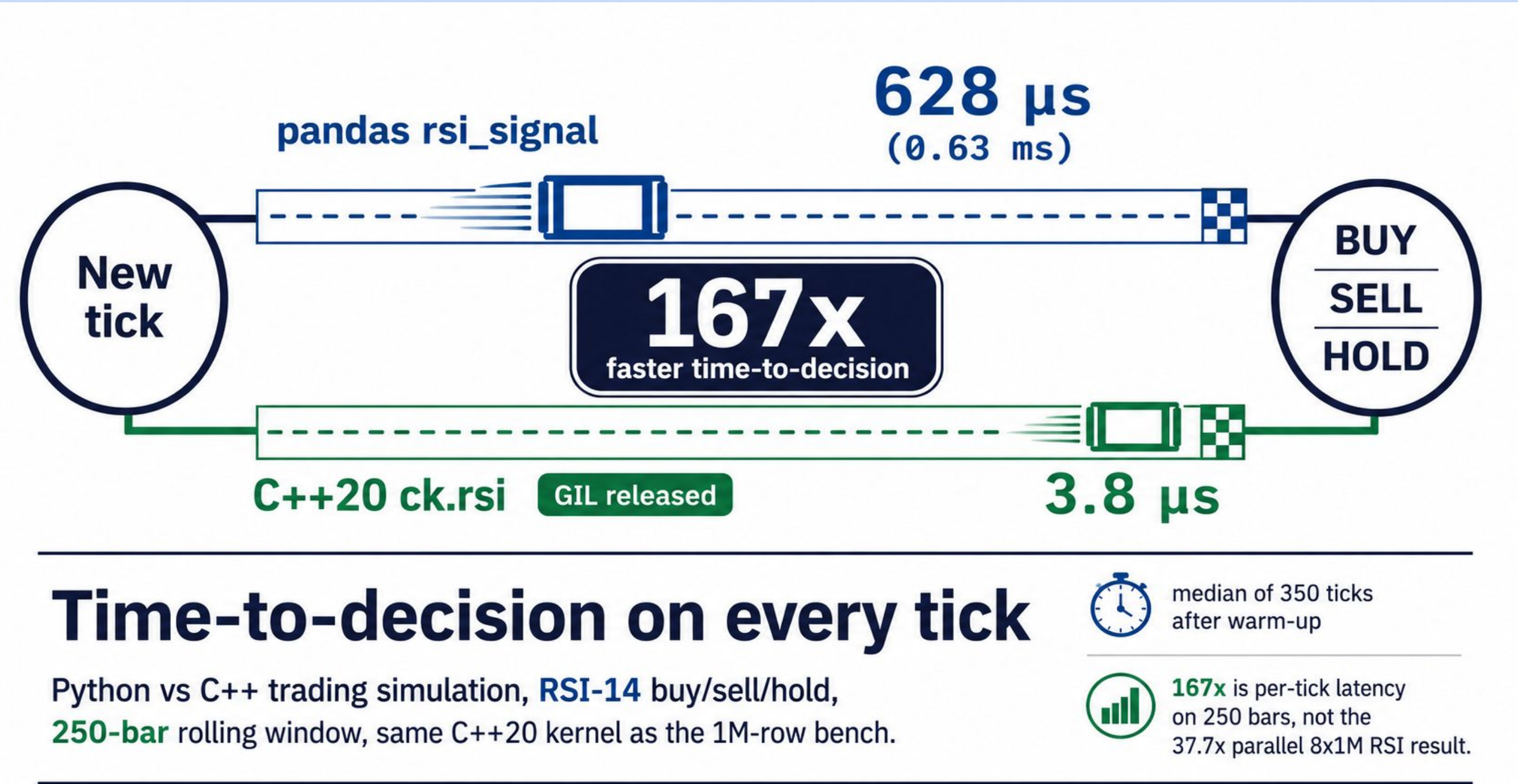


Benchmark Results

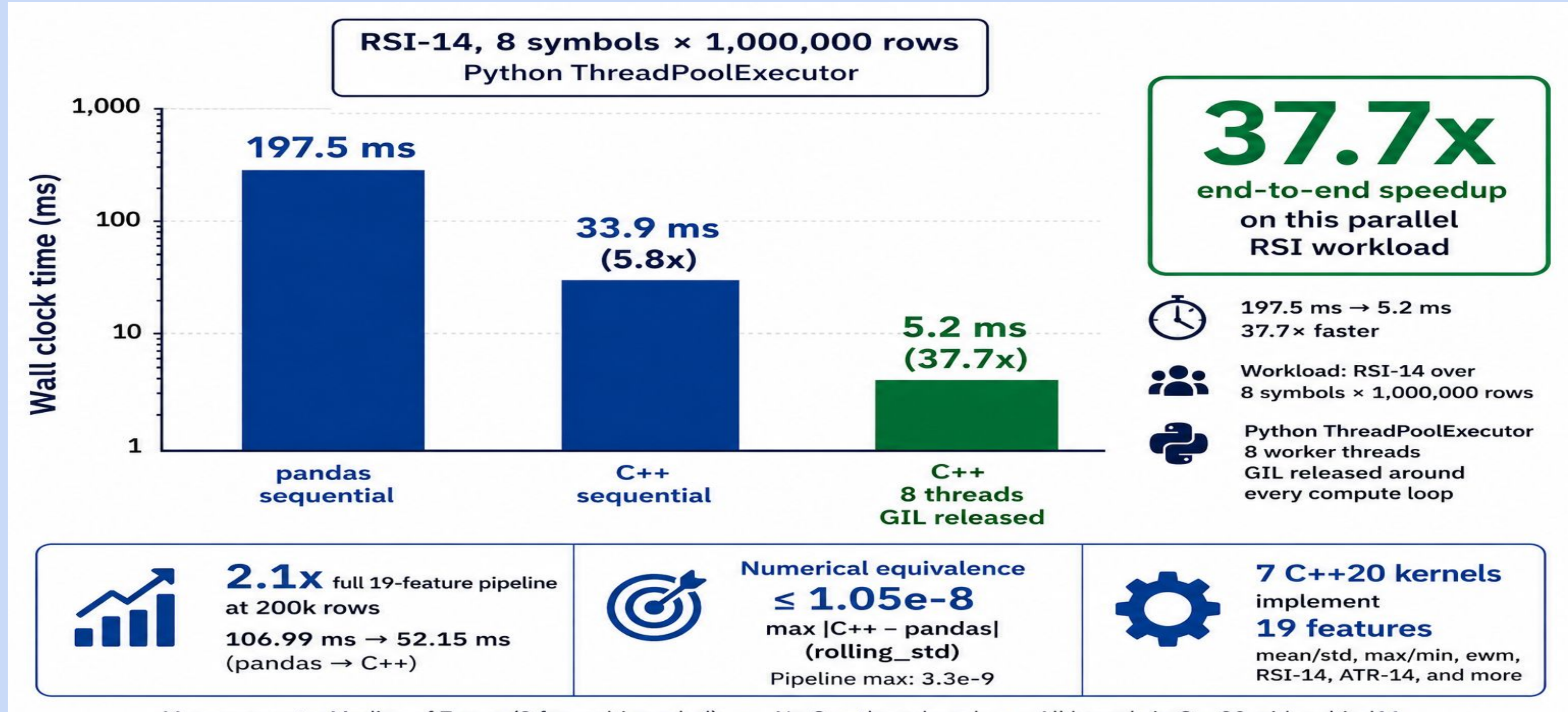
Per-Kernel Speedup by Dataset Size (10k / 100k / 1M Rows)



Python vs C++ Trading Simulation Results



GIL-Release Multi-Symbol RSI Concurrency



AEQUITAS (Agentic Equity & Quantitative Intelligence Trading Analysis System) - Quantitative Research Platform: Live at <https://aequitas-platform.vercel.app/>

